

Week 1 Notes

Natural Numbers, Integers, Rationals & Real Numbers

Course: Mathematics 1 for Data Science

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Lecture 1 — Natural Numbers & Their Operations

1 Numbers: Starting from the Basics

Numbers are abstract concepts used for **counting**. When we see 7 balls and 7 pencils, the number “7” captures what is common — the *quantity*.

Natural Numbers $\mathbb{N}$

Natural Numbers

$$\mathbb{N} = \{0, 1, 2, 3, 4, 5, \dots\}$$

The natural numbers are the counting numbers, **starting from 0**.

Note: Some textbooks define $\mathbb{N} = \{1, 2, 3, \dots\}$ (excluding 0). To emphasize that 0 is included, we sometimes write $\mathbb{N}_0$.

Zero (of Indian origin) is arguably the most important number — without it, our place-value number system would not work.

2 Operations on Natural Numbers

The four basic operations: addition, subtraction, multiplication, division.

Key question: When we perform an operation on two natural numbers, do we always get a natural number back? This property is called **closure**.

Operation	Closed in $\mathbb{N}$?	Why / Why not
Addition	Yes	Sum of two naturals is always natural
Multiplication	Yes	Product of two naturals is always natural
Subtraction	No	$5 - 6 = -1 \notin \mathbb{N}$ (goes below zero)
Division	No	$7 \div 2 = 3.5 \notin \mathbb{N}$ (not an integer)

Subtraction failing closure motivates the extension to **integers**.

3 Integers $\mathbb{Z}$

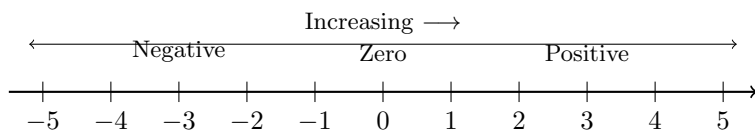
Integers

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

The integers extend the natural numbers by adding **negative numbers** symmetrically around zero.

$\mathbb{N} \subset \mathbb{Z}$ — every natural number is an integer, but not vice versa.

The Number Line



Numbers increase from left to right. The integers extend to $-\infty$ on the left and $+\infty$ on the right.

4 Multiplication: Repeated Addition

Multiplication

$M \times N$ means: take M objects and make N copies of them.

$$M \times N = \underbrace{M + M + M + \dots + M}_{N \text{ times}}$$

Multiplication is repeated addition.

Example: $7 \times 4 = 7 + 7 + 7 + 7 = 28$.

Notation: $M \times N$ can also be written as $M \cdot N$ or simply MN (when using variable names, not digits).

Sign Rules for Multiplication

Signs	Result	Example
$(+) \times (+)$	+	$7 \times 4 = 28$
$(-) \times (+)$	−	$(-7) \times 4 = -28$
$(+) \times (-)$	−	$7 \times (-4) = -28$
$(-) \times (-)$	+	$(-7) \times (-4) = 28$

Rule: Even number of minus signs $\Rightarrow$ positive. Odd number $\Rightarrow$ negative.

5 Exponentiation: Repeated Multiplication

Exponentiation

$$M^K = \underbrace{M \times M \times M \times \cdots \times M}_{K \text{ times}}$$

Exponentiation is repeated multiplication, just as multiplication is repeated addition.

Notation	Means	Name	Geometric interpretation
M^2	$M \times M$	M squared	Area of an $M \times M$ square
M^3	$M \times M \times M$	M cubed	Volume of an $M \times M \times M$ cube
M^K	M multiplied K times	M to the power K	No geometric name for $K > 3$

Hierarchy: Addition $\xrightarrow{\text{repeat}}$ Multiplication $\xrightarrow{\text{repeat}}$ Exponentiation.

6 Division: Repeated Subtraction

Division as Repeated Subtraction

To compute $20 \div 5$: start with 20, keep subtracting 5 until you reach 0.

$$20 \rightarrow 15 \rightarrow 10 \rightarrow 5 \rightarrow 0 \quad (4 \text{ rounds})$$

So $20 \div 5 = 4$.

If the subtraction doesn't reach exactly 0, what's left over is the **remainder**.

Example with remainder: $19 \div 5$:

$$19 \rightarrow 14 \rightarrow 9 \rightarrow 4 \quad (3 \text{ rounds, } 4 \text{ left over})$$

Quotient = 3, **Remainder** = 4.

7 The Modulus (Remainder) Operation

Modulus

$$a \bmod b = \text{remainder when } a \text{ is divided by } b$$

Examples: $19 \bmod 5 = 4$, $20 \bmod 5 = 0$, $7 \bmod 3 = 1$.

8 Divisibility and Factors

Divisibility

a **divides** b (written $a \mid b$) if $b \bmod a = 0$ — that is, a goes into b with no remainder.

Equivalently: $b = k \cdot a$ for some integer k (i.e., b is a **multiple** of a).

Notation: $a \mid b$ means “ a divides b ”. $a \nmid b$ means “ a does not divide b ”.

Examples: $4 \mid 20$ (because $4 \times 5 = 20$), $7 \mid 63$ (because $7 \times 9 = 63$), $32 \mid 1024$ (because $32 \times 32 = 1024$).

$4 \nmid 19$, $9 \nmid 100$.

Factors Come in Pairs

If $a \mid b$ and $b = k \cdot a$, then k is also a factor of b . So factors come in pairs (a, k) where $a \cdot k = b$.

Example — Factors of 12: $(1, 12)$, $(2, 6)$, $(3, 4)$ — six factors total (even count).

Exception — Perfect squares: For $36 = 6 \times 6$, the pair $(6, 6)$ contributes only one distinct factor. So 36 has an **odd** number of distinct factors: $(1, 36)$, $(2, 18)$, $(3, 12)$, $(4, 9)$, (6) .

Rule: Non-square integers have an **even** number of factors. Perfect squares have an **odd** number of factors.

9 Prime Numbers

Prime Number

A prime p is an integer greater than 1 with **exactly two factors**: 1 and p itself.

Note: 1 is **not** a prime (it has only one factor: itself). The smallest prime is **2**.

First few primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, ...

After 2, no even number can be prime (all even numbers are divisible by 2).

Sieve of Eratosthenes

An algorithm to find all primes up to N :

1. List all numbers from 2 to N .
2. The first unmarked number (2) is prime. Mark all its multiples as non-prime.
3. The next unmarked number (3) is prime. Mark all its multiples.
4. Repeat: the next unmarked number is always prime; mark its multiples.
5. Continue until all numbers are either marked prime or marked as multiples.

This systematically generates all primes up to N without missing any.

10 Prime Factorization (Fundamental Theorem of Arithmetic)

Unique Prime Factorization

Every integer $n > 1$ can be written as a product of prime numbers in **exactly one way** (up to the order of factors):

$$n = p_1^{a_1} \cdot p_2^{a_2} \cdot p_3^{a_3} \cdots p_k^{a_k}$$

where $p_1 < p_2 < \cdots < p_k$ are distinct primes and $a_i \geq 1$.

Examples:

- $12 = 2 \times 2 \times 3 = 2^2 \times 3$
- $126 = 2 \times 3 \times 3 \times 7 = 2 \times 3^2 \times 7$
- $1024 = 2^{10}$

Although 12 can be factored as 2×6 or 4×3 or 1×12 , the prime factorization $2^2 \times 3$ is **unique**.

11 Summary

Concept	Key Point
$\mathbb{N} = \{0, 1, 2, \dots\}$	Natural numbers (counting, starts at 0)
$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$	Integers (naturals + negatives)
Multiplication	Repeated addition
Exponentiation	Repeated multiplication
Division	Repeated subtraction; quotient + remainder
$a \bmod b$	Remainder of $a \div b$
$a \mid b$	a divides b (no remainder)
Factors	Come in pairs; odd count $\Leftrightarrow$ perfect square
Prime p	Exactly 2 factors: 1 and p ; smallest is 2
Prime factorization	Every integer has a unique prime decomposition

12 Key Takeaways

1. **Natural numbers** $\mathbb{N} = \{0, 1, 2, \dots\}$ are for counting. **Integers** $\mathbb{Z}$ extend $\mathbb{N}$ with negatives.
2. Subtraction is not closed in $\mathbb{N}$ (can produce negatives), motivating the extension to $\mathbb{Z}$.
3. **Hierarchy:** addition $\rightarrow$ multiplication (repeated addition) $\rightarrow$ exponentiation (repeated multiplication). Division is repeated subtraction.
4. The **modulus** $a \bmod b$ gives the remainder. If the remainder is 0, then $a \mid b$ (a divides b).
5. Factors come in **pairs**. Non-squares have even factor count; perfect squares have odd factor count.
6. **Primes** have exactly two factors (1 and themselves). 1 is not prime; 2 is the smallest prime.
7. The **Sieve of Eratosthenes** generates all primes up to N by iteratively marking multiples.

8. Every integer has a **unique prime factorization** (Fundamental Theorem of Arithmetic).

Lecture 2 — Rational Numbers

1 Why Rational Numbers?

Division is not closed in $\mathbb{Z}$: there is no integer k such that $5k = 19$. To handle such cases, we extend integers to **fractions**.

Example: $19 \div 5 = 3\frac{4}{5}$ — this is a **rational number**.

2 Definition of Rational Numbers $\mathbb{Q}$

Rational Numbers

$$\mathbb{Q} = \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0 \right\}$$

A rational number is a **ratio** of two integers (hence the name “rational”).
 p is the **numerator** (top), q is the **denominator** (bottom).

Hierarchy: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q}$ — every integer n is rational (write it as $n/1$).

3 Non-Unique Representation

Unlike integers (where 7 can only be written as 7), a rational number has **many equivalent representations**:

$$\frac{3}{5} = \frac{6}{10} = \frac{30}{50} = \frac{60}{100} = \dots$$

Multiplying both numerator and denominator by the same nonzero integer gives an **equivalent fraction**: $\frac{p}{q} = \frac{p \cdot k}{q \cdot k}$ for any $k \neq 0$.

Why Non-Uniqueness Is Useful

Addition: To add fractions with different denominators, convert to a **common denominator**:

$$\frac{3}{5} + \frac{3}{4} = \frac{12}{20} + \frac{15}{20} = \frac{27}{20}$$

Comparison: To compare fractions, make denominators equal:

$$\frac{3}{5} < \frac{3}{4} \implies \frac{12}{20} < \frac{15}{20} \implies \frac{3}{5} < \frac{3}{4} \checkmark$$

The common denominator need not be the *smallest* common multiple — any common multiple works.
For instance, using 100: $\frac{60}{100} < \frac{75}{100}$ gives the same conclusion.

4 Reduced Form and GCD

Reduced (Canonical) Form

A fraction $\frac{p}{q}$ is in **reduced form** if p and q share no common factor other than 1, i.e.:

$$\gcd(p, q) = 1$$

To reduce a fraction: divide both numerator and denominator by their **greatest common divisor (GCD)**.

Finding the GCD via Prime Factorization

Example: Reduce $\frac{18}{60}$.

$$18 = 2 \times 3^2, \quad 60 = 2^2 \times 3 \times 5$$

Common prime factors: one 2 and one 3 $\Rightarrow \gcd(18, 60) = 2 \times 3 = 6$.

$$\frac{18}{60} = \frac{18 \div 6}{60 \div 6} = \frac{3}{10}$$

Check: $3 = 3 \times 1$, $10 = 2 \times 5$. No common factors $\Rightarrow$ reduced form. $\checkmark$

GCD Algorithm (via Prime Factorization)

1. Find the prime factorization of both numbers.
2. For each prime, take the **minimum** of the exponents from both factorizations.
3. Multiply these together $\Rightarrow$ GCD.

Note: This is intuitive but not the most efficient method. Euclid's algorithm is faster for large numbers.

5 Density of Rational Numbers

Rationals Are Dense

Between any two distinct rational numbers, there is **always another rational number**.

Proof: Given $\frac{m}{n} < \frac{p}{q}$, their average

$$\frac{1}{2} \left(\frac{m}{n} + \frac{p}{q} \right)$$

is also rational and lies strictly between them.

Since you can repeat this process infinitely, there are **infinitely many** rationals between any two rationals.

Discrete vs. Dense

Property	Integers $\mathbb{Z}$	Rationals $\mathbb{Q}$
“Next” element?	Yes ($m + 1$ follows m)	No
Gap between neighbors?	Yes (nothing between m and $m + 1$)	No (always one in between)
Type	Discrete	Dense

Key contrast: For integers, we can always say “the next one is $m + 1$ ” and “the previous one is $m - 1$ ” with nothing in between. For rationals, there is no such thing as “the next” rational number.

6 Summary

Concept	Key Point
$\mathbb{Q} = \{p/q : p, q \in \mathbb{Z}, q \neq 0\}$	Rationals = ratios of integers
Non-unique representation	$3/5 = 6/10 = 30/50 = \dots$
Common denominator	Needed for addition and comparison
Reduced form	$\gcd(p, q) = 1$; use prime factorization
$\gcd(18, 60) = 6$	Common primes: $2 \times 3 = 6$
Dense	Always a rational between any two rationals
Discrete vs. dense	$\mathbb{Z}$ is discrete, $\mathbb{Q}$ is dense

7 Key Takeaways — Lecture 2

1. Division not being closed in $\mathbb{Z}$ motivates the extension to **rational numbers** $\mathbb{Q} = \{p/q\}$.
2. Rational numbers have **non-unique** representations ($p/q = pk/qk$). This is useful for finding common denominators.
3. **Addition and comparison** of fractions require a common denominator (any common multiple of the two denominators works).
4. The **reduced form** (canonical representation) has $\gcd(p, q) = 1$. Find the GCD via prime factorization and divide both numerator and denominator.
5. Rationals are **dense**: between any two rationals there is always another (take their average). There is no “next” rational number.
6. Integers are **discrete**: $m + 1$ is the next integer after m , with nothing in between.
7. Number system hierarchy: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q}$ — each extension fixes a closure failure (subtraction, then division).

Lecture 3 — Real & Complex Numbers

1 Do Rationals Fill the Number Line?

The rationals $\mathbb{Q}$ are dense — between any two rationals there is another rational. Does this mean all points on the number line are rational?

No. There are “gaps” in $\mathbb{Q}$ that only become visible when we consider operations like the square root.

2 Square Roots and Perfect Squares

Square Root

The **square root** of M is the number r such that $r \times r = M$.

For **perfect squares** $(1, 4, 9, 16, 25, \dots)$, the square root is an integer:

$$\sqrt{1} = 1, \quad \sqrt{4} = 2, \quad \sqrt{9} = 3, \quad \sqrt{16} = 4, \quad \sqrt{25} = 5, \quad \sqrt{256} = 16$$

Question: What about $\sqrt{M}$ when M is *not* a perfect square?

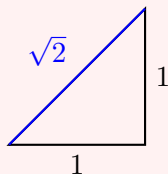
Example: $\sqrt{10}$ lies between 3 and 4 (since $3^2 = 9$ and $4^2 = 16$). Is it rational?

3 Irrational Numbers

$\sqrt{2}$ Is Irrational

$\sqrt{2}$ **cannot** be written as p/q for any integers p and q .

Yet $\sqrt{2}$ is a perfectly real, measurable quantity: by Pythagoras’s theorem, the diagonal of a unit square has length $\sqrt{1^2 + 1^2} = \sqrt{2}$.



So $\sqrt{2}$ is on the number line but is **not rational** — it is an **irrational number**.

General rule: For any integer M that is *not* a perfect square, $\sqrt{M}$ is irrational. This includes $\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{7}, \sqrt{8}, \sqrt{10}, \dots$

4 Famous Irrational Numbers

Number	Approx. value	Where it appears
π	3.14159265...	Ratio of circumference to diameter of <i>any</i> circle
e	2.71828182...	Base of natural logarithms
$\sqrt{2}$	1.41421356...	Diagonal of a unit square

All three have **infinite, non-repeating** decimal expansions (a hallmark of irrational numbers).

5 Real Numbers $\mathbb{R}$

Real Numbers

$$\mathbb{R} = \text{Rational numbers} \cup \text{Irrational numbers}$$

The **real numbers** $\mathbb{R}$ include every point on the number line — all rationals *and* all irrationals.

Density: Like $\mathbb{Q}$, the real numbers are **dense**. Between any two reals r and r' ($r < r'$), the average $(r + r')/2$ is a real number strictly between them.

6 Complex Numbers (Preview)

Why Do We Need Yet More Numbers?

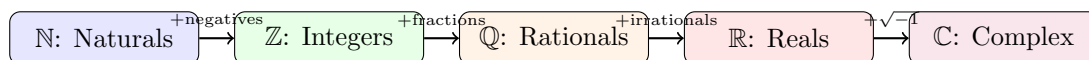
Problem: $\sqrt{-1}$ has no real solution.

By the sign rule for multiplication: $(+) \times (+) = (+)$ and $(-) \times (-) = (+)$. Any number multiplied by itself gives a **positive** result. So no real number squared gives -1 .

Solution: Define a new number $i = \sqrt{-1}$ and build the **complex numbers** $\mathbb{C}$, which extend $\mathbb{R}$.

(Complex numbers are not needed for this course.)

7 The Complete Number System Hierarchy



$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$$

Each extension is motivated by an operation that **fails** in the previous system: subtraction ($\mathbb{N} \rightarrow \mathbb{Z}$), division ($\mathbb{Z} \rightarrow \mathbb{Q}$), square roots of non-squares ($\mathbb{Q} \rightarrow \mathbb{R}$), square roots of negatives ($\mathbb{R} \rightarrow \mathbb{C}$).

Discrete vs. Dense

Number system	Discrete or Dense?	“Next” exists?
$\mathbb{N}$ (naturals)	Discrete	Yes ($n + 1$)
$\mathbb{Z}$ (integers)	Discrete	Yes ($m + 1$)
$\mathbb{Q}$ (rationals)	Dense	No (take average)
$\mathbb{R}$ (reals)	Dense	No (take average)

8 Key Takeaways — Lecture 3

1. **Irrational numbers** are real numbers that cannot be written as p/q . Example: $\sqrt{2}$, $\sqrt{3}$, π , e .

2. $\sqrt{M}$ is irrational for every non-perfect-square integer M .
3. $\sqrt{2}$ is geometrically real (diagonal of a unit square) but algebraically not rational.
4. $\mathbb{R} = \mathbb{Q} \cup \{\text{irrationals}\}$ — the real numbers fill the **entire** number line.
5. Real numbers are **dense** (same argument as rationals: take the average).
6. **Complex numbers** $\mathbb{C}$ extend $\mathbb{R}$ to allow $\sqrt{-1}$, but are not needed for this course.
7. **Complete hierarchy:** $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$. Each extension fixes a missing operation.

Lecture 4 — Introduction to Sets

1 What Is a Set?

A **set** is a collection of items (called **elements** or **members**). Sets are written with curly braces, elements separated by commas.

Examples:

- Days of the week: $\{\text{Sun, Mon, Tue, Wed, Thu, Fri, Sat}\}$ (7 elements).
- Factors of 24: $\{1, 2, 3, 4, 6, 8, 12, 24\}$ (8 elements).
- Primes below 15: $\{2, 3, 5, 7, 11, 13\}$ (6 elements).
- $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ — all infinite sets of numbers.

Key properties of sets:

- **No ordering:** $\{\text{Kohli, Dhoni, Pujara}\} = \{\text{Pujara, Kohli, Dhoni}\}$.
- **No duplicates:** Adding Kohli a second time does not change the set.
- **Mixed types allowed:** A set can contain people, instruments, animals, numbers — anything.

2 Notation: Element Of ($\in$)

Membership

$x \in X$ means “ x is an element of set X ”.

$x \notin X$ means “ x is not an element of set X ”.

Examples: $0 \in \mathbb{N}$, $5 \in \mathbb{Z}$, $\sqrt{2} \notin \mathbb{Q}$.

3 Cardinality

The **cardinality** of a set is the number of elements it contains, written $|X|$.

For finite sets, just count. For infinite sets, measuring cardinality is more subtle (a separate topic).

Examples: $|\{\text{factors of 24}\}| = 8$. $|\mathbb{N}| = \infty$.

Surprising Finite Sets: Platonic Solids

Regular polygons (2D) are infinite in number (triangle, square, pentagon, ... for any number of sides). But their 3D counterparts — **Platonic solids** (where every face is an identical regular polygon meeting at equal angles) — number exactly **5**:

Solid	Faces	Face shape
Tetrahedron	4	Triangle
Cube	6	Square
Octahedron	8	Triangle
Dodecahedron	12	Pentagon
Icosahedron	20	Triangle

4 Subsets

Subset

$X \subseteq Y$ means **every element of X is also an element of Y** .

Proper subset: $X \subset Y$ (or $X \subsetneq Y$) means $X \subseteq Y$ and $X \neq Y$.

Examples:

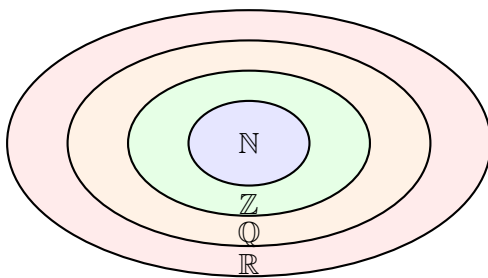
- $\{\text{Kohli, Pujara}\} \subset \{\text{Kohli, Dhoni, Pujara}\}$.
- $\text{Primes} \subset \mathbb{N}$ (not all naturals are prime).
- $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$ (proper subsets at each level).

Two important facts:

1. Every set is a subset of itself: $X \subseteq X$ (trivially, every element of X is in X).
2. **Set equality:** $X = Y$ if and only if $X \subseteq Y$ and $Y \subseteq X$.

Venn Diagrams

Sets and subset relationships are visualized using **Venn diagrams** — circles (or closed curves) where one circle inside another indicates a subset.



5 The Empty Set

Empty Set $\emptyset$

The **empty set** $\emptyset$ is the set with **no elements**. It is the “zero” of set theory.

$\emptyset$ is a **subset of every set**: $\emptyset \subseteq X$ for all X .

Why? The statement “every element of $\emptyset$ is in X ” is *vacuously true* — there are no elements to contradict it. (Like saying “all 3-legged birds have pink beaks” — true because no 3-legged birds exist.)

Important distinction: $\emptyset \neq \{\emptyset\}$.

- $\emptyset$ has 0 elements.
- $\{\emptyset\}$ has 1 element (namely, the empty set itself).

6 Power Set

Power Set

The **power set** of X , written $\mathcal{P}(X)$ or 2^X , is the set of **all subsets** of X .
If $|X| = n$, then $|\mathcal{P}(X)| = 2^n$.

Example: $X = \{a, b\}$. Subsets: $\emptyset, \{a\}, \{b\}, \{a, b\}$. So $\mathcal{P}(X) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$, with $|\mathcal{P}(X)| = 2^2 = 4$.

$\mathcal{P}(\emptyset) = \{\emptyset\}$ (one element: the empty set itself). $|\mathcal{P}(\emptyset)| = 2^0 = 1$.

Why 2^n Subsets?

Argument 1 (Choices): For each of the n elements, choose to **include** or **exclude** it from the subset. That gives $2 \times 2 \times \cdots \times 2 = 2^n$ possible subsets.

Argument 2 (Binary numbers): Each subset corresponds to an n -bit binary string, where bit i is 1 if x_i is included, 0 if excluded.

Example: $X = \{a, b, c, d\}$. The binary string 0110 means: exclude a , include b , include c , exclude d $\Rightarrow$ subset $\{b, c\}$.

Binary	Decimal	Subset of $\{a, b, c, d\}$
0000	0	$\emptyset$
0001	1	$\{d\}$
0101	5	$\{b, d\}$
1111	15	$\{a, b, c, d\}$

There are exactly $2^4 = 16$ four-bit binary numbers (0000 to 1111), so exactly 16 subsets.

7 A Note on Russell's Paradox

Can we collect *all* sets into one “set of all sets”? Bertrand Russell showed this leads to a **contradiction** (Russell's Paradox). In practice, for the collections of numbers and objects we use in this course, this is not a concern.

8 Key Takeaways — Lecture 4

1. A **set** is an unordered collection of distinct elements. Notation: $\{a, b, c\}$.
2. $x \in X$ means x belongs to X ; $x \notin X$ means it does not.
3. **Cardinality** $|X|$ = number of elements. Can be finite or infinite.
4. $X \subseteq Y$ (subset): every element of X is in Y . $X \subsetneq Y$ (proper subset): $X \subseteq Y$ and $X \neq Y$.
5. $X = Y$ iff $X \subseteq Y$ and $Y \subseteq X$.
6. The **empty set** $\emptyset$ has no elements and is a subset of every set (vacuous truth).
7. $\emptyset \neq \{\emptyset\}$: the first has 0 elements, the second has 1.

8. The **power set** $\mathcal{P}(X)$ contains all subsets of X . If $|X| = n$, then $|\mathcal{P}(X)| = 2^n$.
9. The 2^n count follows from: each element has 2 choices (include/exclude), or equivalently, subsets biject with n -bit binary strings.

Lecture 5 — Set Comprehension, Intervals & Set Operations

1 Set Comprehension: Defining Subsets of Infinite Sets

We cannot list all elements of an infinite set. Instead, we describe subsets using a **condition** applied to a known set — this is called **set comprehension**.

Set Comprehension Notation

$$\{x \in S \mid \text{condition on } x\}$$

Read as: “the set of all x in S such that [condition] holds.”

Three parts:

1. **Starting set** S (e.g., $\mathbb{Z}$, $\mathbb{N}$, $\mathbb{R}$) — we must begin from a known set.
2. **Condition** (the filter) — decides which elements to keep.
3. **Collector** $\{x \mid \dots\}$ — gathers all elements satisfying the condition.

Examples

Even integers:

$$\{x \in \mathbb{Z} \mid x \bmod 2 = 0\}$$

Take all integers, keep only those divisible by 2.

Perfect squares:

$$\{m \in \mathbb{N} \mid \sqrt{m} \in \mathbb{N}\}$$

Take all natural numbers, keep only those whose square root is also a natural number. Gives $\{1, 4, 9, 16, 25, \dots\}$.

Rationals in reduced form:

$$\left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0, \gcd(p, q) = 1 \right\}$$

Keep only fractions where numerator and denominator share no common factor other than 1.

2 Intervals: Subsets of $\mathbb{R}$

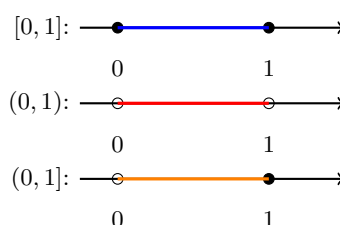
Intervals describe **ranges of real numbers** between two endpoints. They are defined using set comprehension with inequality conditions.

Four Types of Intervals

Name	Notation	Endpoints	Definition
Closed	$[a, b]$	Both included	$\{r \in \mathbb{R} \mid a \leq r \leq b\}$
Open	(a, b)	Both excluded	$\{r \in \mathbb{R} \mid a < r < b\}$
Left-open	$(a, b]$	Left excluded	$\{r \in \mathbb{R} \mid a < r \leq b\}$
Right-open	$[a, b)$	Right excluded	$\{r \in \mathbb{R} \mid a \leq r < b\}$

Convention: Square bracket $[]$ = endpoint **included**. Round bracket $()$ = endpoint **excluded**.

Visualizing on the Number Line



Filled circle = included. Open circle = excluded.

Practical importance: The interval $[0, 1]$ (or $(0, 1)$) appears frequently in probability, where values range from 0 to 1.

3 Set Operations

Union ($\cup$)

Union

$X \cup Y$ = all elements in X **or** Y (or both). Duplicates appear only once.

$\{a, b, c\} \cup \{c, d, e\} = \{a, b, c, d, e\}$

Note: $|X \cup Y| \leq |X| + |Y|$ (equality only if X and Y share no elements).

Union is **commutative**: $X \cup Y = Y \cup X$.

Intersection ($\cap$)

Intersection

$X \cap Y$ = all elements in **both** X and Y .

$\{a, b, c, d\} \cap \{a, d, e, f\} = \{a, d\}$

Intersection is **commutative**: $X \cap Y = Y \cap X$.

Set Difference ($\setminus$ or $-$)

Set Difference

$X \setminus Y$ (or $X - Y$) = all elements in X that are **not** in Y .

$$\{a, b, c, d\} \setminus \{a, d, e, f\} = \{b, c\}$$

$$\{a, d, e, f\} \setminus \{a, b, c, d\} = \{e, f\} \quad (\text{different!})$$

Set difference is **not commutative**: $X \setminus Y \neq Y \setminus X$ in general.

Example: $\mathbb{R} \setminus \mathbb{Q}$ = all irrational numbers. $\mathbb{Q} \setminus \mathbb{Z}$ = all non-integer rationals.

Complement ($\overline{X}$ or X^c)

Complement

The **complement** of X is everything **not** in X — but with respect to what?

A complement is only meaningful relative to a specified **universe** U :

$$\overline{X} = U \setminus X$$

Example: Primes $\subset \mathbb{N}$. If $U = \mathbb{N}$, then the complement of the primes = composite numbers $\cup \{0, 1\}$.

If $U = \mathbb{R}$, the complement would include $\pi, e, \sqrt{2}, \dots$ — not what we intend!

Rule: Always specify the universe when using complement.

4 Summary of Set Operations

Operation	Symbol	Commutative?	Result
Union	$X \cup Y$	Yes	Elements in X or Y
Intersection	$X \cap Y$	Yes	Elements in both X and Y
Set difference	$X \setminus Y$	No	Elements in X but not Y
Complement	$\overline{X}$	N/A	$U \setminus X$ (needs universe U)

5 Key Takeaways — Lecture 5

1. **Set comprehension** $\{x \in S \mid \text{condition}\}$ defines subsets of infinite sets by specifying a starting set and a filter condition.
2. Examples: even integers ($x \bmod 2 = 0$), perfect squares ($\sqrt{m} \in \mathbb{N}$), rationals in reduced form ($\gcd(p, q) = 1$).
3. **Intervals** are subsets of $\mathbb{R}$ between two endpoints. Square bracket = included, round bracket = excluded. Four types: $[a, b]$, (a, b) , $(a, b]$, $[a, b)$.
4. **Union** ($\cup$): combines elements from both sets; commutative.
5. **Intersection** ($\cap$): keeps only common elements; commutative.
6. **Set difference** ($\setminus$): removes elements of the second set from the first; **not** commutative.

7. **Complement:** everything not in X , relative to a specified **universe** U . Always state the universe.

Lecture 5A — Set Comprehension: Examples & Practice

1 Quick Review

- **Membership:** $x \in X$ (element belongs to set). **Subset:** $X \subseteq Y$ (every element of X is in Y).
- **Empty set** $\emptyset$ is a subset of every set. Every set is a subset of itself.
- **Power set:** $|\mathcal{P}(X)| = 2^{|X|}$.
- **Set comprehension:** $\{f(x) \mid x \in S, \text{condition}(x)\}$ builds new sets from old ones.

2 Anatomy of Set Comprehension

Three Components

$$\{ \underbrace{x^2}_{\text{Transform}} \mid \underbrace{x \in \mathbb{Z}}_{\text{Generator}}, \underbrace{x \bmod 2 = 0}_{\text{Filter}} \}$$

1. **Generator** — draws elements from an existing set ($x \in \mathbb{Z}$).
 2. **Filter** — keeps only elements satisfying a condition (x is even).
 3. **Transform** — converts the surviving elements into the final output ($x \rightarrow x^2$).
- If no filter is needed, omit it. If no transform is needed (identity), just keep x as-is.

Worked Example: Squares of Even Integers

$$\{x^2 \mid x \in \mathbb{Z}, x \bmod 2 = 0\}$$

Step 1 (Generate): All integers: $\dots, -3, -2, -1, 0, 1, 2, 3, \dots$

Step 2 (Filter): Keep only even: $\dots, -4, -2, 0, 2, 4, \dots$

Step 3 (Transform): Square each: $(-4)^2 = 16, (-2)^2 = 4, 0^2 = 0, 2^2 = 4, 4^2 = 16, \dots$

Step 4 (Remove duplicates): $(-2)^2 = 2^2 = 4$, so keep only one copy.

Result: $\{0, 4, 16, 36, 64, \dots\}$

3 Examples with Identity Transform

When the transform is just “keep as-is,” we write x (not $f(x)$) on the left side.

Rationals in reduced form:

$$\left\{ \frac{p}{q} \mid \frac{p}{q} \in \mathbb{Q}, \gcd(p, q) = 1 \right\}$$

Generator: all rationals. Filter: $\gcd = 1$. Transform: identity (keep p/q as-is).

Half-open interval $[-1, 2)$:

$$\{r \in \mathbb{R} \mid -1 \leq r < 2\}$$

Generator: all reals. Filter: between -1 (included) and 2 (excluded). Transform: identity.

4 Replacing a Long List with Set Comprehension

Task: Cubes of the first 500 natural numbers.

Tedious way: $\{0^3, 1^3, 2^3, \dots, 499^3\}$ — must list all 500 numbers explicitly.

Compact way using set comprehension:

Define an intermediate set: $X = \{n \in \mathbb{N} \mid n < 500\}$

Then: $\{n^3 \mid n \in X\}$

This is readable, precise, and avoids the informal “...” notation.

5 Multiple Ways to Define the Same Set

Perfect Squares: Three Equivalent Definitions

Version 1 (Filter on integers):

$$\{z \in \mathbb{Z} \mid \sqrt{z} \in \mathbb{Z}\}$$

Generate all integers, keep those whose square root is also an integer.

Version 2 (Filter on naturals):

$$\{m \in \mathbb{N} \mid \sqrt{m} \in \mathbb{N}\}$$

Since squares are always ≥ 0 , we can restrict to $\mathbb{N}$ instead of $\mathbb{Z}$.

Version 3 (Transform, no filter):

$$\{n^2 \mid n \in \mathbb{N}\}$$

Generate all naturals, square each one. No filter needed — every natural squared is a perfect square.

All three produce the same set: $\{0, 1, 4, 9, 16, 25, \dots\}$

Key insight: A filter in one formulation can become a transform in another. The generator set matters — changing from $\mathbb{Z}$ to $\mathbb{N}$ is valid here because squares are non-negative.

6 Extending to Other Number Systems

The concept of “perfect square” extends beyond integers:

Rational perfect squares:

$$\{q^2 \mid q \in \mathbb{Q}\} \quad \text{e.g., } \left(\frac{3}{4}\right)^2 = \frac{9}{16}$$

Not every rational is a perfect square in $\mathbb{Q}$. For example, $1/2$ is not, because $\sqrt{1/2} = 1/\sqrt{2}$ is irrational.

Lesson: The same set comprehension structure applies across different number systems — just change the generator set. But the resulting sets will differ depending on the domain.

7 Key Takeaways — Lecture 5A

1. Set comprehension has three parts: **generator** (source set), **filter** (condition), and **transform** (output function).
2. The transform can produce duplicates (e.g., $(-2)^2 = 2^2$), which are automatically removed (sets have no duplicates).
3. If no transform is needed, use the **identity** (keep the element as-is).
4. Set comprehension replaces tedious explicit listings with compact, precise descriptions — essential for infinite sets.
5. The **same set** can often be defined in multiple equivalent ways by swapping filters and transforms or changing the generator.
6. Perfect squares can be defined either by filtering (keep if $\sqrt{m} \in \mathbb{N}$) or by transforming (square every $n \in \mathbb{N}$).
7. Changing the generator (e.g., $\mathbb{Z} \rightarrow \mathbb{Q}$) extends the concept to other number systems, but the resulting set changes.

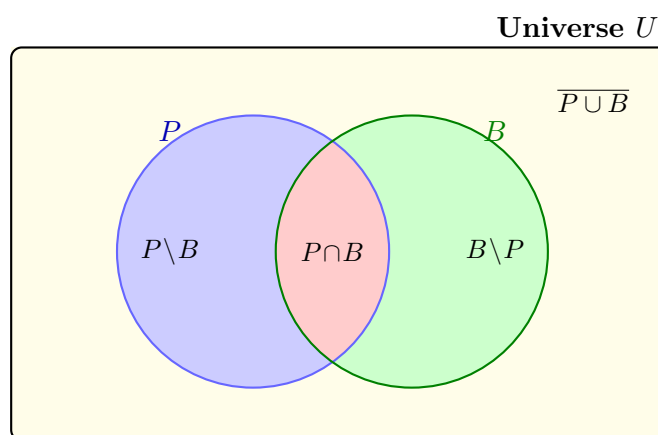
Lecture 5B — Set Operations & Counting with Venn Diagrams

1 Recap: Set Operations via Venn Diagrams

Operation	Symbol	Venn diagram region
Union	$X \cup Y$	Entire shaded area (both circles)
Intersection	$X \cap Y$	Overlapping region only
Set difference	$X \setminus Y$	Left circle minus overlap
Complement	$\overline{X}$	Everything outside X (needs universe)

2 The Four Disjoint Regions

When two sets P and B overlap within a universe U , the Venn diagram creates **four disjoint regions** that together account for every element:



$$|U| = |P \setminus B| + |P \cap B| + |B \setminus P| + |\overline{P \cup B}|$$

These four regions are **mutually exclusive** (no element appears in two regions) and **exhaustive** (every element of U is in exactly one region).

3 Counting Problems with Venn Diagrams

Problem 1: Find the Total

Problem

In a class: 30 students took Physics, 25 took Biology, 10 took both, 5 took neither. How many students are in the class?

Solution:

- $|P \cap B| = 10$ (both subjects).

- $|P \setminus B| = 30 - 10 = 20$ (Physics only).
- $|B \setminus P| = 25 - 10 = 15$ (Biology only).
- $|\overline{P \cup B}| = 5$ (neither).

$$|U| = 20 + 10 + 15 + 5 = 50 \text{ students}$$

Problem 2: Find a Missing Region

Problem

Class has 55 students. 32 took Physics, 11 took both Physics and Biology, 7 took neither. How many took Biology but not Physics?

Solution:

- $|P \cap B| = 11$.
- $|P \setminus B| = 32 - 11 = 21$.
- $|\overline{P \cup B}| = 7$.
- Let $|B \setminus P| = x$.

$$7 + 21 + 11 + x = 55 \implies x = 16 \text{ students took Biology but not Physics}$$

Problem 3: Find the Intersection

Problem

Class has 60 students. 35 took Physics, 30 took Biology, 10 took neither. How many took both?

Solution:

Students taking at least one subject: $|P \cup B| = 60 - 10 = 50$.

If there were no overlap, the total would be $|P| + |B| = 35 + 30 = 65$. But only 50 spots exist, so $65 - 50 = 15$ students were **double-counted** (they take both).

$$|P \cap B| = |P| + |B| - |P \cup B| = 35 + 30 - 50 = 15$$

This is the **inclusion-exclusion principle** for two sets:

$$|P \cup B| = |P| + |B| - |P \cap B|$$

Verification: $|P \setminus B| = 35 - 15 = 20$, $|B \setminus P| = 30 - 15 = 15$. Total: $20 + 15 + 15 + 10 = 60$. ✓

4 The Inclusion-Exclusion Principle

Inclusion-Exclusion (Two Sets)

$$|A \cup B| = |A| + |B| - |A \cap B|$$

When we add $|A|$ and $|B|$, elements in both sets are counted **twice**. Subtracting $|A \cap B|$ corrects for the double-counting.

This formula connects all four quantities: $|A|$, $|B|$, $|A \cup B|$, $|A \cap B|$. Given any three, solve for the fourth.

5 Summary of the Week

Topic	Lecture
Natural numbers, integers, operations, primes	L1
Rational numbers, GCD, reduced form, density	L2
Irrational numbers, reals, hierarchy $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$	L3
Sets, subsets, cardinality, power set (2^n)	L4
Set comprehension, intervals, set operations	L5
Set comprehension examples (generator/filter/transform)	L5A
Venn diagram counting, inclusion-exclusion	L5B

6 Key Takeaways — Lecture 5B

1. A Venn diagram for two overlapping sets creates **four disjoint regions**: $P \setminus B$, $P \cap B$, $B \setminus P$, and $\overline{P \cup B}$.
2. The total is the sum of all four regions: $|U| = |P \setminus B| + |P \cap B| + |B \setminus P| + |\overline{P \cup B}|$.
3. **Three problem types**: find the total, find a missing region, find the intersection — all solved by populating the four regions and using the total equation.
4. The **inclusion-exclusion principle**: $|A \cup B| = |A| + |B| - |A \cap B|$ corrects for double-counting.
5. Venn diagrams are not just illustrations — they are **computational tools** for solving counting problems systematically.

Lecture 6 — Relations

Cartesian Products, Properties & Equivalence Relations

1 Cartesian Product

Cartesian Product

The **Cartesian product** $A \times B$ is the set of all **ordered pairs** (a, b) where $a \in A$ and $b \in B$:

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

Order matters: $(0, 1) \neq (1, 0)$.

$$|A \times B| = |A| \cdot |B|.$$

Example: $A = \{0, 1\}$, $B = \{2, 3\}$:

$$A \times B = \{(0, 2), (0, 3), (1, 2), (1, 3)\} \quad |A \times B| = 2 \times 2 = 4$$

Visualization: For number sets like $\mathbb{N} \times \mathbb{N}$ or $\mathbb{R} \times \mathbb{R}$, the product is a **grid** or **plane**. Each pair (a, b) is a point with x -coordinate a and y -coordinate b .

Extension to n -tuples

The product extends to any number of sets: $A_1 \times A_2 \times \cdots \times A_n$ produces **n -tuples** $(a_1, a_2, \dots, a_n)$.

Example — Pythagorean triples: $(a, b, c) \in \mathbb{N} \times \mathbb{N} \times \mathbb{N}$ such that $a, b, c > 0$ and $a^2 + b^2 = c^2$.

E.g., $(3, 4, 5)$, $(5, 12, 13)$, $(8, 15, 17)$.

2 Relations

Binary Relation

A **relation** R on sets A and B is a **subset** of $A \times B$:

$$R \subseteq A \times B$$

If $(a, b) \in R$, we say “ a is related to b ” and write $a R b$.

A relation = Cartesian product + set comprehension (filter out the pairs of interest).

Examples of Relations

Successor relation on $\mathbb{N}$: $\{(m, n) \in \mathbb{N} \times \mathbb{N} \mid n = m + 1\} = \{(0, 1), (1, 2), (2, 3), \dots\}$

Divisibility on $\mathbb{N}$: $\{(d, n) \in \mathbb{N} \times \mathbb{N} \mid d \mid n\}$ — e.g., $(2, 82)$, $(14, 56)$.

Circle of radius 5: $\{(a, b) \in \mathbb{R} \times \mathbb{R} \mid a^2 + b^2 = 25\}$ — the set of all points at distance 5 from the origin. This defines a geometric shape (circle) as a relation.

Teacher–course allocation: $A \subseteq T \times C$ where T = teachers, C = courses. Pairs $(t, c) \in A$ mean teacher t teaches course c . Visualized as a bipartite graph with arrows.

Mother-child: $\{(m, c) \in P \times P \mid m \text{ is the mother of } c\}$ where P is a set of people.

3 The Identity Relation

$$I_A = \{(a, a) \mid a \in A\}$$

Every element is paired with **itself**. On $\mathbb{N} \times \mathbb{N}$, this is the diagonal: $(0, 0), (1, 1), (2, 2), \dots$

4 Properties of Binary Relations

For a relation $R \subseteq A \times A$ (both elements from the *same* set):

Four Key Properties

Property	Definition	Example
Reflexive	$(a, a) \in R$ for all a	Divisibility: $a \mid a$ ✓
Symmetric	$(a, b) \in R \Rightarrow (b, a) \in R$	$\gcd(a, b) = 1$: order doesn't matter ✓
Transitive	$(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$	<: if $3 < 10$ and $10 < 28$ then $3 < 28$ ✓
Anti-symmetric	$(a, b) \in R$ and $a \neq b \Rightarrow (b, a) \notin R$	<: if $a < b$ then $b \not< a$ ✓

Note: Anti-symmetry $\neq$ “not symmetric.” Anti-symmetry says: if a pair is present, the reverse must be *absent* (for distinct elements). It does not require every pair to be present.

5 Equivalence Relations

Equivalence Relation

A relation R is an **equivalence relation** if it is simultaneously:

1. **Reflexive:** $a R a$ for all a .
2. **Symmetric:** $a R b \Rightarrow b R a$.
3. **Transitive:** $a R b$ and $b R c \Rightarrow a R c$.

An equivalence relation behaves like **equality**: it groups elements that are “the same” in some sense.

Example: Integers Modulo 5

Define $a \sim b$ iff $(b - a) \bmod 5 = 0$ (i.e., a and b differ by a multiple of 5).

Check the properties:

- Reflexive: $(a - a) = 0$, and $0 \bmod 5 = 0$. ✓
- Symmetric: if $(b - a) \bmod 5 = 0$, then $(a - b) \bmod 5 = 0$. ✓

- Transitive: if $b - a$ and $c - b$ are multiples of 5, so is $c - a = (c - b) + (b - a)$. ✓

6 Equivalence Classes and Partitions

Key Result

Every equivalence relation **partitions** the set into disjoint **equivalence classes**:

- All elements within a class are equivalent to each other.
- No element belongs to two different classes.
- The classes together cover the entire set.

Example: $\mathbb{Z} \bmod 5$ partitions the integers into 5 classes:

Remainder	Class members
0	$\dots, -10, -5, 0, 5, 10, 15, \dots$
1	$\dots, -9, -4, 1, 6, 11, 16, \dots$
2	$\dots, -8, -3, 2, 7, 12, 17, \dots$
3	$\dots, -7, -2, 3, 8, 13, 18, \dots$
4	$\dots, -6, -1, 4, 9, 14, 19, \dots$

Real-world example: A 12-hour clock partitions 24 hours into 2 equivalence classes (AM/PM). 2:00 AM and 2:00 PM look the same on the clock — they are “equivalent mod 12.”

7 Key Takeaways — Lecture 6

1. The **Cartesian product** $A \times B$ is the set of all ordered pairs (a, b) . Order matters. $|A \times B| = |A| \cdot |B|$.
2. A **relation** $R \subseteq A \times B$ is a subset of the Cartesian product — selected pairs satisfying some condition.
3. Relations model geometric shapes (circles), allocations (teacher–course), family ties, divisibility, and more.
4. Extends to n -tuples: Pythagorean triples, square corners, etc.
5. Four key properties: **reflexive** ($a R a$), **symmetric** ($a R b \Rightarrow b R a$), **transitive** (chain rule), **anti-symmetric** (no reverse pairs).
6. An **equivalence relation** (reflexive + symmetric + transitive) partitions a set into disjoint **equivalence classes** that behave like equality.
7. Example: $\mathbb{Z} \bmod 5$ creates 5 equivalence classes based on remainder. A 12-hour clock is mod12 equivalence.

Lecture 7 — Functions

Domain, Codomain, Range & Properties of Mappings

1 What Is a Function?

Function

A **function** is a rule that assigns to each input **exactly one** output.

Notation: $f : X \rightarrow Y$ means f maps elements of the **domain** X to the **codomain** Y .

For a specific input: $x \mapsto f(x)$. Example: $\text{sq}(x) = x^2$.

Two requirements:

1. **Defined everywhere:** For every $x \in X$, there exists a y such that $f(x) = y$.
2. **Unique output:** For each x , there is exactly **one** $y = f(x)$.

2 Domain, Codomain, and Range

Three Sets Associated with Every Function

Set	Definition	Example ($f(x) = x^2, f : \mathbb{R} \rightarrow \mathbb{R}$)
Domain	Set of all valid inputs	$\mathbb{R}$ (all reals)
Codomain	Set of <i>possible</i> outputs	$\mathbb{R}$ (all reals)
Range	Set of <i>actual</i> outputs	$[0, \infty) = \{r \in \mathbb{R} \mid r \geq 0\}$

Key: $\text{Range} \subseteq \text{Codomain}$. They may or may not be equal.

3 Functions as Relations

Every function $f : X \rightarrow Y$ defines a relation:

$$R_f = \{(x, y) \in X \times Y \mid y = f(x)\}$$

Plotting a function = plotting the relation R_f . The graph of $f(x) = x^2$ is the set of all points (x, x^2) — the parabola.

What distinguishes a function from a general relation:

- Every x in the domain has *at least one* pair (function is defined everywhere).
- Every x has *at most one* pair (output is unique).

4 Examples of Functions

Linear function: $f(x) = mx + c$

- m = **slope** (angle/steepness). c = **intercept** (where line crosses y -axis).
- Same slope, different intercept $\Rightarrow$ parallel lines.

- Negative slope $\Rightarrow$ line goes downward.
- Domain = $\mathbb{R}$, codomain = $\mathbb{R}$, range = $\mathbb{R}$ (line spans all values).

Square root: $f(x) = \sqrt{x}$

- By convention, $\sqrt{x}$ means the **positive** square root.
- If we took both $+\sqrt{x}$ and $-\sqrt{x}$, the output would not be unique $\Rightarrow$ not a function!
- Domain depends on codomain: if codomain = $\mathbb{R}$, then domain = $[0, \infty)$ (no real square root of negatives). If codomain = $\mathbb{C}$, domain can be all of $\mathbb{R}$.

5 Properties of Functions

Injective (One-to-One)

Injective

f is **injective** if distinct inputs always produce distinct outputs:

$$x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$$

Injective: $f(x) = 3.5x + 5.7$ (different $x \rightarrow$ different point on the line). ✓

Not injective: $f(x) = x^2$ (because $f(a) = f(-a)$ for all a). ✗

Surjective (Onto)

Surjective

f is **surjective** if every element of the codomain is actually hit:

$$\text{Range} = \text{Codomain} \iff \forall y \in Y, \exists x \in X : f(x) = y$$

Surjective: $f(x) = mx + c$ with $f : \mathbb{R} \rightarrow \mathbb{R}$ (line spans all reals). ✓

Not surjective: $f(x) = 5x^2 + 3$ with codomain $\mathbb{R}$ (range = $[3, \infty) \neq \mathbb{R}$). ✗

Not surjective: $f(x) = 7\sqrt{x}$ with codomain $\mathbb{R}$ (only non-negative outputs). ✗

Bijjective (One-to-One and Onto)

Bijjective = Injective + Surjective

f is **bijjective** if it is both injective and surjective:

- Every input maps to a **distinct** output (injective).
- Every output has a **unique** input that maps to it (surjective).

A bijection establishes a **one-to-one correspondence** between domain and codomain.

Theorem: f is bijjective $\iff f$ is injective *and* surjective.

Example: $f(x) = 3.5x + 5.7$ with $f : \mathbb{R} \rightarrow \mathbb{R}$ is bijjective (injective line that spans all reals).

6 Summary of Function Properties

Function	Injective?	Surjective?	Bijjective?
$f(x) = mx + c \ (\mathbb{R} \rightarrow \mathbb{R})$	Yes	Yes	Yes
$f(x) = x^2 \ (\mathbb{R} \rightarrow \mathbb{R})$	No	No	No
$f(x) = \sqrt{x} \ ([0, \infty) \rightarrow \mathbb{R})$	Yes	No	No
$f(x) = \sqrt{x} \ ([0, \infty) \rightarrow [0, \infty))$	Yes	Yes	Yes

Note: Whether a function is surjective (and hence bijective) depends on the choice of codomain!

7 Bijections and Cardinality

Bijections for Counting

Two sets A and B have the **same cardinality** ($|A| = |B|$) if and only if there exists a **bijection** $f : A \rightarrow B$.

For **finite sets**: this is a convenience (like pairing marbles from two sacks one-by-one; both sacks empty together $\Rightarrow$ same count).

For **infinite sets**: this is the *only* way to compare cardinalities.

Example: Every line $y = mx + c$ is uniquely determined by the pair $(m, c) \in \mathbb{R} \times \mathbb{R}$. This defines a bijection between lines and points in $\mathbb{R}^2$. Therefore, the number of lines equals the number of points on the plane.

Caution: Two points determine a unique line, but many pairs of points determine the *same* line. So the map from pairs-of-points to lines is **not** injective — it is **not a bijection**. You cannot conclude $|\mathbb{R}^2 \times \mathbb{R}^2| = |\mathbb{R}^2|$ from this.

8 Key Takeaways — Lecture 7

1. A **function** assigns exactly one output to each input. Notation: $f : X \rightarrow Y, x \mapsto f(x)$.
2. **Domain** = valid inputs. **Codomain** = possible outputs. **Range** = actual outputs. $\text{Range} \subseteq \text{Codomain}$.
3. Functions correspond to relations $R_f = \{(x, f(x))\}$. Plotting a function = plotting this relation.
4. **Injective** (one-to-one): distinct inputs $\rightarrow$ distinct outputs.
5. **Surjective** (onto): range = codomain. Every output is reachable.
6. **Bijjective** = injective + surjective. Establishes one-to-one correspondence.
7. Two sets have the **same cardinality** iff a bijection exists between them — the only way to compare infinite sets.
8. Whether a function is surjective/bijjective **depends on the codomain**. Changing the codomain can change the classification.

Lecture 7A — Relations as Tables & the Join Operation

1 Cartesian Products Revisited

Recall: $A \times B = \{(a, b) \mid a \in A, b \in B\}$. Order matters: $A \times B \neq B \times A$. Size: $|A \times B| = |A| \cdot |B|$.

Self-product: $B \times B$ includes all pairs — not just (b, b) but also (b_1, b_2) where $b_1 \neq b_2$.

Example: $A = \{1, 4, 7\}$, $B = \{1, 16, 49\}$. $A \times B$ has $3 \times 3 = 9$ pairs. A relation $S \subseteq A \times B$ picking only $(1, 1), (4, 16), (7, 49)$ is the “squaring” relation ($b = a^2$).

Beyond pairs: $A \times B \times A$ gives **triples**. In general, n sets give n -**tuples**.

2 Relations on Numbers: More Examples

Divisibility

$$D = \{(d, n) \in \mathbb{N} \times \mathbb{N} \mid d \mid n\}$$

Contains $(7, 63)$, $(17, 85)$, etc. Extending the generator from $\mathbb{N}$ to $\mathbb{Z}$ adds negative divisors (e.g., $(-7, 63)$).

Prime Powers

First define primes: $\text{Primes} = \{p \in \mathbb{N} \mid p \neq 1, \text{factors}(p) = \{1, p\}\}$.

Then: $\text{PrimePowers} = \{(p, n) \in \text{Primes} \times \mathbb{N} \mid n = p^m \text{ for some } m \in \mathbb{N}\}$

Contains $(5, 625)$ since $5^4 = 625$, and $(3, 1)$ since $3^0 = 1$ by definition.

3 Relations Beyond Numbers: Airlines Example

Let C = set of cities served by an airline. The **direct flights** relation:

$$D \subseteq C \times C, \quad D = \{(a, b) \in C \times C \mid \text{direct flight from } a \text{ to } b\}$$

Properties:

- **Irreflexive:** No flight from a city to itself. $(a, a) \notin D$ for all a .
- **Not necessarily symmetric:** Triangular routes exist (e.g., Chennai $\rightarrow$ Madurai direct, but return via Salem).

4 Tables Are Relations

Key Insight: Tables = Relations

A **table** (as in a database or spreadsheet) is a relation. Each **row** is a tuple; each **column** draws values from a set.

Example — Flight distances:

Source	Destination	Distance (km)
Bangalore	Chennai	290
Chennai	Delhi	1752
Delhi	Mumbai	1148

This table is a relation on $C \times C \times \mathbb{N}$.

Practical benefit of knowing properties:

- Distance is **symmetric**: Chennai–Delhi = Delhi–Chennai = 1752 km. So we only need to store **half** the entries.
- Distance to self is always 0 (**reflexive** in a trivial sense): no need to store those rows.

5 Keys and Key-Value Pairs

Keys

A **key** is a column (or set of columns) whose value **uniquely identifies** each row.

Example — Student table:

Roll No. (key)	Name	DOB
CS2101	Abhay	2003-05-12
CS2102	Pai Gauche	2002-11-03
CS2103	Pai Gauche	2003-01-17

Names can repeat (two “Pai Gauche”), but roll numbers are unique. Given a roll number, you can find exactly one row — this makes the table behave like a **function** (roll number $\mapsto$ student record).

6 The Join Operation

Join: Combining Two Tables on a Common Column

Given two tables (relations) that share a common column (e.g., roll number), the **join** merges rows where the shared column values **match**.

Students table: (Roll, Name, DOB)

Grades table: (Roll, Subject, Grade)

Joined result: (Roll, Name, Subject, Grade)

$$\{(r, n, s, g) \mid (r, n, d) \in \text{Students}, (r', s, g) \in \text{Grades}, r = r'\}$$

Only rows with **matching roll numbers** are combined. The key ensures the correct name is paired with the correct grades — even when two students share the same name.

Example result:

Roll	Name	Subject	Grade
CS2101	Abhay	Maths	A
CS2102	Pai Gauche	Maths	A
CS2103	Pai Gauche	Physics	B

The two “Pai Gauche” entries are correctly distinguished by their different roll numbers.

7 Why Relations Matter for Data Science

Concept	Data Science Application
Table = Relation	Databases store data as relations
Key = Unique identifier	Prevents ambiguity (e.g., duplicate names)
Join = Combine tables	Merge data from multiple sources
Symmetric relation	Store only half the entries (save space)
Reflexive/irreflexive	Identify trivial or impossible entries

8 Key Takeaways — Lecture 7A

1. Cartesian products are ordered ($A \times B \neq B \times A$), extend to n -tuples, and include self-products ($A \times A$).
2. Relations are subsets of Cartesian products, defined explicitly or via set comprehension.
3. **Tables are relations.** Every database table, spreadsheet, or data frame is a relation on a Cartesian product of column value sets.
4. A **key** uniquely identifies each row — makes the table behave like a function ($\text{key} \mapsto \text{record}$).
5. The **join** combines two tables on matching values in a shared column. It is the fundamental operation for merging data.

6. Knowing relation properties has practical value: symmetric relations need only half the storage; irreflexive relations exclude self-pairs.

Lecture 7B — Function Examples: Ranges & Growth

1 Recap: Functions on Numbers

A function $f : X \rightarrow Y$ maps each input $x \in X$ to exactly one output $f(x) \in Y$.

Domain = valid inputs. **Codomain** = possible outputs. **Range** = actual outputs ($\subseteq$ codomain).

Functions beyond numbers: The “mother” function maps every person to their (unique) mother. But in this course, we focus on **functions on numbers**.

2 Question 1: What Is the Range?

Even when the codomain is all of $\mathbb{R}$, the range may be smaller.

Function	Range	Why
$f(x) = x^2$	$[0, \infty)$	Squaring always gives ≥ 0 ; no upper bound
$f(x) = x^3 - 3x^2 + 5$	$(-\infty, \infty)$	Cubic grows to $\pm\infty$
$f(x) = 5 \sin x$	$[-5, +5]$	Sine oscillates in $[-1, 1]$; scaled by 5

Bounded vs. unbounded:

- x^2 : unbounded above, bounded below (by 0).
- $x^3 - 3x^2 + 5$: unbounded in both directions.
- $5 \sin x$: bounded in both directions (range is a closed interval).

3 Question 2: Where Are the Maxima and Minima?

Global vs. Local Extrema

Global maximum/minimum: The largest/smallest value the function *ever* takes.

Local maximum/minimum: A point where the function is larger/smaller than its *immediate neighbours* (the function “turns around”).

Function	Global min	Global max	Local features
x^2	0 (at $x = 0$)	None ($\rightarrow \infty$)	Single minimum
$x^3 - 3x^2 + 5$	None ($\rightarrow -\infty$)	None ($\rightarrow \infty$)	Local max at $x = 0$, local min at $x = 2$
$5 \sin x$	-5	+5	Periodic; infinitely many

The cubic example: $f(x) = x^3 - 3x^2 + 5$ has no global extrema (grows without bound), but it zigzags: rises to a **local maximum** at $x = 0$, dips to a **local minimum** at $x = 2$, then rises again. Finding these turning points is a key topic in calculus.

The sine example: $5 \sin x$ attains its global max (+5) and global min (-5) **infinitely often**, periodically. The local and global extrema coincide.

4 Question 3: How Fast Does a Function Grow?

Comparing Growth Rates

Given two functions, which one **eventually dominates**?

$x^3 - 3x^2 + 5$ vs. x^2 : Initially x^2 may be larger, but eventually the cubic overtakes the quadratic and stays above it forever.

General principle: Higher-degree polynomials grow faster. x^3 eventually beats x^2 , which eventually beats x .

Why Growth Rates Matter: A Data Science Example

Let $g(y)$ = number of data science graduates in year y .

Let $j(y)$ = number of new data science jobs in year y .

- If $j(y)$ grows faster than $g(y)$: demand exceeds supply $\rightarrow$ more students enrol.
- If $g(y)$ grows faster than $j(y)$: supply exceeds demand $\rightarrow$ graduates struggle to find jobs.

Comparing growth rates of real-world quantities is a core data analysis task.

5 Summary: Three Key Questions About Functions

Question	What we want to know
Range	What output values can the function actually produce?
Extrema	Where does the function reach its highest/lowest values (global or local)?
Growth rate	How fast does the function increase, and which function dominates?

These questions will be studied in detail using calculus tools (derivatives, limits) in later weeks.

6 Key Takeaways — Lecture 7B

1. The **range** of a function may be much smaller than the codomain. x^2 has range $[0, \infty)$ even though codomain is $\mathbb{R}$.
2. Functions can be **bounded** ($5 \sin x \in [-5, 5]$) or **unbounded** ($x^3 \rightarrow \pm\infty$).
3. **Global extrema** are the overall max/min. **Local extrema** are turning points where the function reverses direction.
4. A cubic like $x^3 - 3x^2 + 5$ has local max at $x = 0$ and local min at $x = 2$, but no global extrema.
5. **Growth rate comparison:** higher-degree polynomials eventually dominate lower-degree ones (x^3 beats x^2).
6. Comparing growth rates has real-world applications: jobs vs. graduates, revenue vs. costs, supply vs. demand.

References & Acknowledgements

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